

WORKSHEET

The Tic-Tac-Toe World: Agents, Problems, and Performance

A second simple environment, like the vacuum world — but this one is multi-agent and adversarial, and its legal actions change as the game progresses. You play X; the opponent plays O. Work through this in your group.

Name(s): _____

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Part 1 — The Agent Function

For each board below, the percept is the full board itself (tic-tac-toe is fully observable, so percept = state). Write the move a rational X-agent should make.

Board A

X to move. Your move (row, col): _____

Board B

X	X	
O		
	O	

X to move. Your move (row, col): _____

Board C

O	O	
X		
	X	

X to move. Your move (row, col): _____

Board D

X	O	
		X
O		

X to move. Your move (row, col): _____

Discuss: in the vacuum world, the action set {Left, Right, Suck} was the same in every state. Is that true here? What determines which actions are legal in a given tic-tac-toe state?

Part 2 — Formalizing Tic-Tac-Toe as a Game

Fill in each part of the formal problem definition. Note: the last row asks for a utility, not a path cost — think about why a step-counting cost doesn't fit this environment the way it fit the vacuum world.

State space	
Initial state	
Actions	
Transition model	
Terminal test	
Utility (at terminal states)	

Part 3 — Choosing a Performance Measure

Consider three candidate performance measures for the X-agent:

A	+1 point every time the agent's mark occupies the center or a corner square
B	+1 point at the end of the game for a win; 0 for a draw or a loss
C	+1 for a win, -1 for a loss, 0 for a draw, evaluated only once the game ends

For each measure: what behavior does it actually reward, and can an agent score well without playing well?

	What does it reward?	Can it be gamed? How?	Use it?
A			
B			
C			

Reflection

Measure B treats a loss the same as a draw. Why might that be a worse flaw here than it would be in a single-agent environment like the vacuum world?